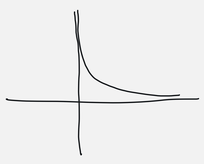
$$\int_0^{\infty} f(x)\,dx = \lim_{x\to+\infty} \int_0^{x} f(t)\,dt$$

$(u_n)_n$. $\forall \varepsilon>0, \exists n_0 \mid \forall n \geq n_0, \ |u_n - L| \leq \varepsilon$

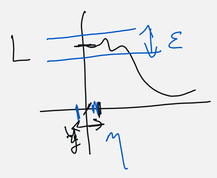
$\forall A>0 \qquad u_n > A$

$f \xrightarrow[x\to+\infty]{} L$: $\forall \varepsilon>0, \exists x_0 \mid \forall x \geq x_0, \ |f(x)-L| \leq \varepsilon$

$]x_0, +\infty[$

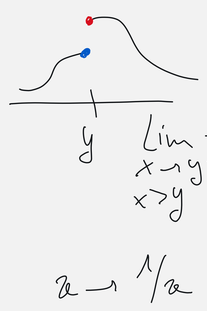


$\lim_{x\to y} f(x)$: $\forall \varepsilon>0, \exists \eta>0 \mid 0<|x-y|<\eta \Rightarrow |f(x)-L| \leq \varepsilon$



$f(x) = \frac{1}{x \log^{\alpha}(x)}$

$\int_0^{c} f(t)\,dt = \lim_{\substack{X\to 0 \\ X>0}} \int_X^{c} f(t)\,dt$



$\lim_{\substack{x\to y \\ x>y}} f(x)$ $\lim_{\substack{x\to y \\ x<y}} f(x)$



$x \to 1/x$

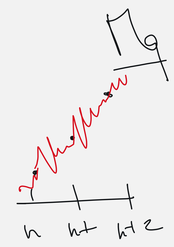


$\lim_{\substack{u\to 0 \\ u>0}} 1/x = +\infty$

$\lim_{\substack{u\to 0 \\ u<0}} 1/u = -\infty$

$\int_{\cdot}^{+\infty} f(t)\,dt$ $\int_{-\infty}^{+\infty}$ / $\int_0^{1} \frac{1}{x(x-1)}\,dx$ $\int_{\varepsilon}^{1/2} \frac{1}{x(x-1)}\,dx \xrightarrow[\substack{\varepsilon\to 0 \\ \varepsilon>0}]{}$, $\int_{1/2}^{1-\varepsilon} \frac{1}{x(x-1)}\,dx \xrightarrow[\substack{\varepsilon\to 0 \\ \varepsilon>0}]{}$

Ex 1: a - $a>0$. $f: [a,+\infty[\to \mathbb{R}$, $f(t)\geq 0$, $F(x) = \int_a^{x} f(t)\,dt$



F creciente , $\left\{ \int_a^{\infty} f(t)\,dt \ cv \right\} \Longleftrightarrow \{ F \text{ acotada arriba} \}$

$x \leq y \Rightarrow F(x) \leq F(y)$

$F(y) - F(x) = \int_a^{y} f - \int_a^{x} f = \int_x^{y} f$, $f \geq 0 \Rightarrow \int_{\alpha}^{\beta} f \geq 0$ $\int_{\alpha}^{\beta} f = 0$

$F(y) - F(x) \geq 0$ / ≥ 0



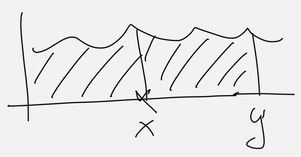
$[\alpha,\beta]$



Si $f=0$



$\Rightarrow F(y) \geq F(x)$



$\Leftarrow$ F está acotada en $\mathbb{R}$ $([a,+\infty))$. $(a_n)_n$ sec. y $\leq \Pi$ necesariamente tiene que cv

$\forall \varepsilon>0 \ \exists n_0 \mid n \geq n_0 \Rightarrow |a_n - L| \leq \varepsilon$

$\exists L$ tq $\forall \varepsilon>0, \exists x_0 \in \mathbb{R} \mid \forall x \geq x_0, \ |F(x) - L| \leq \varepsilon$

$\left(F(x) \xrightarrow[x\to+\infty]{} \sup_{[a,+\infty)} F \right)$

$\Rightarrow$

(a,b) , $[a,b]$, $[a,b[$, $]a,b]$, $]a,b[$

$[a,b)$, $[a,b)$, $(a,b]$, (a,b)

Si F no es acotada arriba, pues es creciente

$\forall n, \exists x_n \in \mathbb{R}$ tq $F(x_n) > n$

y $x_{n+1} > x_n$ $(x_n, +\infty)$

$\int_a^{x_n} f = F(x_n) \xrightarrow[n\to+\infty]{} +\infty$

$> n$

$\int_a^{x_n} f \xrightarrow[n\to+\infty]{} +\infty$. Si $\int_a^{x} f(t)\,dt \xrightarrow[x\to+\infty]{} L \in \mathbb{R}$

$\Rightarrow \varphi(n)$, $\int_a^{\varphi(n)} f \xrightarrow[n\to+\infty]{} L$